COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 9 — Requirements Engineering Process

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Section 13: Classroom Activity — Requirements Discovery Exercise

Pair Simulation: Requirements analyst conducting a 15-minute stakeholder interview with a student representative for the College Event Management System.

1. Problem & Stakeholders

- **Core Problem:** Students frequently miss technical competitions due to fragmented notice board announcements, and popular workshops overbook within minutes without transparent queue handling.
- **Stakeholders:** Student Attendee, Student Event Head, Faculty Activity Coordinator.

2. Elicitation Interview (5 Core Questions)

- **Q1:** "How do you currently discover upcoming events on campus, and what information is typically missing?"
- **Q2:** "What specific steps do you take when trying to register for an event that has limited seating?"
- **Q3:** "If an event is completely full, what behavior do you expect from the registration system?"
- **Q4:** "How would you prefer to receive registration confirmations and last-minute schedule updates?"
- **Q5:** "What prevents students from cancelling registrations when their schedule changes?"

3. Elicited Requirements & Refinement

- **FR-01:** The system shall allow students to subscribe to notification tags based on technical domains (e.g., AI, Robotics, Web).
- **FR-02:** The system shall place students on an automated waitlist when event capacity is reached.
- **FR-03:** The system shall automatically promote the first waitlisted student if an active registrant cancels.
- **NFR-01:** The waitlist promotion email must be dispatched within 60 seconds of a cancellation.
- **NFR-02:** Student personal phone numbers must not be visible to student volunteers or other attendees.
- **Vague Statement:** "The notification system should be fast."
- **Measurable Refinement:** "The system shall dispatch transactional confirmation emails within 3 seconds of database commit for 99% of registrations under peak test load."

Section 16: Homework — Requirements Engineering Deliverables Packet

Comprehensive requirements baseline document for the College Event Management System (CEMS).

1. System Scope & Objectives

The CEMS project will centralize event publishing, student registrations, automated ticketing, and attendance validation across all campus departments. Scope boundaries: In-Scope includes event cataloging, role-based workflows, capacity enforcement, QR generation, and attendance reporting. Out-of-Scope includes paid ticketing payment gateway handling (Phase 2) and physical venue door sensor hardware.

2. Stakeholder Needs Register

Stakeholder	Role	Primary Concern / Need
Student Body	Primary consumer	Fair access to high-demand workshops with immediate registration proof.
Faculty Mentors	Event approvers	Ensuring events do not clash with academic exam timetables.
Student Coordinators	Operational managers	Accurate headcount for food catering, badges, and workshop kit distribution.
College Security	Gate control	Preventing unregistered campus outsiders from entering restricted halls.
Dean of Student Affairs	Executive sponsor	Comprehensive end-of-semester participation metrics for accreditation audit.

3. Requirements Catalogue & Traceability

Req ID	Type	Requirement Statement	Source Stakeholder	Acceptance Test Link
REQ-01	FR	Authenticate users via Institutional Google Workspace OAuth2.	IT Administrator	TC-AUTH-01 (OAuth Flow)
REQ-02	FR	Restrict single student registration to 1 seat per event.	Student Coordinators	TC-REG-03 (Duplicate Check)
REQ-03	FR	Auto-close registration when current count equals seat limit.	Faculty Mentors	TC-CAP-02 (Boundary Test)
REQ-04	FR	Generate dynamic digital QR ticket on mobile screen.	Security Staff	TC-TKT-01 (QR Render Test)
REQ-05	NFR	System UI must conform to WCAG	Dean Student Affairs	TC-A11Y-01 (Screen Reader)

		2.1 AA accessibility guidelines.		

4. Change Request Management Scenarios

- **Scenario A (Waitlist Request):** Department heads request automated waitlist promotion. Impact: Requires adding a `WAITLISTED` status to database enum, updating notification workers, and revising registration state machines.
- **Scenario B (Multi-Day Attendance):** Robotics workshop head requests marking attendance separately for Day 1 and Day 2. Impact: Changes attendance schema from 1:1 event relationship to 1:Many session entity model.

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